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GeoV2V: Geometry-Grounded Video Diffusion Model for Driving Scene Generation

LiDAR-Warped Reference Frames and Synthetic Para4D Data Enable Zero-Shot Lane-Shift Video Generation on Real-World Waymo and nuScenes Benchmarks

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Autonomous driving simulators require realistic video from camera poses that were never recorded—a lane shift, a slightly different heading, or a hypothetical trailing vehicle. GeoV2V addresses this challenge by combining LiDAR and depth-based geometric priors with full-video conditioning on the Wan 2.1 video diffusion backbone, generating temporally consistent, geometrically accurate driving videos under large trajectory shifts. Trained entirely on Para4D, a new synthetic CARLA-based dataset of synchronized multi-lane videos, GeoV2V transfers zero-shot to real-world Waymo and nuScenes data, achieving state-of-the-art visual quality and geometric consistency.

AT A GLANCE

Problem: Large viewpoint shifts in novel-view driving video reveal occluded regions; pure generation loses geometric coherence across frames.

Why it matters: High-fidelity novel-view synthesis is essential for scalable simulation of rare events (lane changes, near-misses) that cannot be captured by fixed sensor rigs.

Method: Geometric warping via LiDAR/depth priors + Wan 2.1 video diffusion conditioned on warped sequence + Para4D synthetic training data.

Result: State-of-the-art FID, FVD, and depth-consistency on Waymo and nuScenes without fine-tuning on real data.

Evidence: Waymo, nuScenes, and Para4D benchmarks; ablations isolating geometric conditioning and synthetic pretraining.

THE BIG PICTURE

The long tail of driving scenarios—rare lane changes, unexpected obstacles, adverse weather—cannot be covered by fixed-route data collection. Novel view synthesis (NVS) addresses this by generating plausible videos from camera poses never recorded.

GeoV2V grounds a powerful video diffusion prior (Wan 2.1) with explicit 3D geometry, turning the generation problem into geometry-aware inpainting: warp what is visible, inpaint what is occluded, and preserve temporal dynamics.

WHY EXISTING APPROACHES FALL SHORT

Image-based NVS (NeRF, 3DGS): Require dense multi-view captures unavailable in single-camera drive-by settings.

Pure video generation (Sora, Gen-3): Unconstrained generation ignores 3D structure, producing temporally inconsistent lane layouts.

Warping-only methods: Geometric warp leaves holes where occluded content is revealed; filling them with inpainting lacks temporal coherence.

CORE MECHANISM

GeoV2V operates in three stages: (1) Geometric warping: LiDAR point clouds project each reference frame to the target pose, producing a warped reference sequence with explicit holes where geometry is occluded. (2) Conditional video generation: Wan 2.1 video diffusion transformer takes the warped sequence as conditioning via cross-attention, filling holes and refining artifacts. (3) Para4D training: The model trains on synthetic CARLA multi-lane videos covering diverse weather, lighting, and traffic conditions.

TECHNICAL FORMULATION

Warp operator W using depth D_t and camera intrinsics K:

I_tilde_t = W(I_t^ref, D_t, P_t^ref -> P_t^tgt, K)

Video diffusion model G_theta generates:

V_out = G_theta(z_T, V_tilde, M)

where z_T is Gaussian noise and M are per-pixel visibility masks.

Training objective: L = E[||G_theta(z_t, V_tilde, M) - V_gt||^2] on Para4D pairs.

LEARNING OR INFERENCE PROCEDURE

Training: GeoV2V trains entirely on synthetic Para4D data with lateral offsets of 1.0–5.0 m, covering standard and extreme viewpoints. Wan 2.1 weights are initialised from a pretrained checkpoint.

Inference (zero-shot): On Waymo and nuScenes, only LiDAR-based depth is used for warping; the generator runs without any domain-specific fine-tuning.

WHAT THE GUARANTEE SAYS

The geometric warping step provides an explicit constraint: visible pixels in the reference frame are reproduced at the correct target location with sub-pixel registration accuracy (verified by LiDAR reprojection error). The diffusion model only generates content in provably occluded regions (mask M_t = 0), giving a formal separation between geometric reliability and generative uncertainty.

EXPERIMENTAL FINDINGS

Waymo lane-shift NVS: State-of-the-art FID and FVD against prior methods (WoVoGen, GeoFlow, AnyScene), with >15% FVD improvement at 3.5 m lane shifts.

nuScenes: Consistent gains in geometric depth consistency (homography error reduction >20%) and visual quality metrics.

Challenging lighting: GeoV2V maintains FVD gains in night and adverse weather where geometric priors are noisiest.

Para4D ablation: Replacing Para4D with a warping-only baseline increases FVD by 38%, confirming the importance of explicit cross-lane supervision.

ABLATIONS AND INTERPRETATION

LiDAR vs. monocular depth: LiDAR improves depth consistency by 12%; monocular depth is sufficient for visual quality but not geometric accuracy.

Conditioning mechanism: Replacing cross-attention with concatenation reduces FVD by 8%, confirming that temporal context is best exploited through attention.

Training data scale: 10k Para4D pairs is sufficient; doubling to 20k improves FVD by only 2%.

REFERENCE

Yuchen Xi, Yuhan Guo, Chenwei Hou, Zixu Liu, Jin Fang, Jason Liu, Ruigang Yang. **GeoV2V: Geometry-Grounded Video Diffusion Model for Driving Scene Generation.** arXiv:2610.25245. ECCV 2026. https://arxiv.org/abs/2610.25245